API Endpoint Plan

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user account with a designated role (Organiser or Participant).	None	{ firstName, lastName, email, password, phone, roleId }	201 Created - User registered successfully. 400 Bad Request - Validation error or duplicate email.
POST	/api/auth/login	Authenticates a user and returns an authentication token.	None	{ email, password }	200 OK - Auth token returned. 401 Unauthorized - Invalid credentials.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/profile	Retrieves the profile details of the currently authenticated user.	Any	None	200 OK - User profile object. 401 Unauthorized - Unauthenticated.
PUT	/api/users/profile	Updates the profile information of the currently authenticated user.	Any	{ firstName, lastName, phone }	200 OK - Profile updated. 400 Bad Request - Invalid data.
GET	/api/events	Lists all upcoming events across all categories and types.	None	None	200 OK - Array of events.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}	Retrieves detailed information for a specific event.	None	None	200 OK - Event details. 404 Not Found - Event does not exist.
POST	/api/events	Creates a new event managed by the authenticated organiser.	Organiser	{ name, description, date, location, distance, eventType }	201 Created - Event created. 403 Forbidden - Not an organiser.
PUT	/api/events/{id}	Updates existing event details.	Organiser	{ name, description, date, location, distance, eventType }	200 OK - Event updated.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					404 Not Found - Event not found.
DELETE	/api/events/{id}	Removes an event and associated categories/enrolments.	Organiser	None	200 OK - Event deleted. 404 Not Found - Event not found.
GET	/api/events/{eventId}/categories	Lists all categories defined for a specific event.	None	None	200 OK - Array of categories. 404 Not Found - Event not found.
POST	/api/events/{eventId}/categories	Adds a new age/distance category to an existing event.	Organiser	{ categoryName, minAge, maxAge }	201 Created - Category added.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					400 Bad Request - Invalid age ranges.
POST	/api/enrolments	Enrols the logged-in participant into an event category.	Participant	{ categoryId }	201 Created - Enrolment completed. 409 Conflict - Participant already enrolled.
GET	/api/events/{eventId}/enrolments	Fetches all participant enrolments for a given event.	Organiser	None	200 OK - List of enrolled participants.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					403 Forbidden - Not event owner.
POST	/api/results	Records finishing times and positions for an enrolled participant.	Organiser	{ enrolmentId, finishTime, overallPosition, categoryPosition }	201 Created - Result logged. 404 Not Found - Enrolment ID invalid.
Get	/api/results/my results	Retrieves the personal race history and results for the authenticated participant	Participant	None	200 Ok – List of Historical race results 401 Unauthorized.
GET	/api/events/{eventide}/route	Retrieves route map coordinates, elevation profile and water point for the event.	Any	None	200 Ok – Route GIS and elevation data. 404 Not Found- Route Unavailable